Accelerated Marsh Erosion Following the *Deepwater Horizon* Oil Spill Confirmed, Ameliorated by Planting

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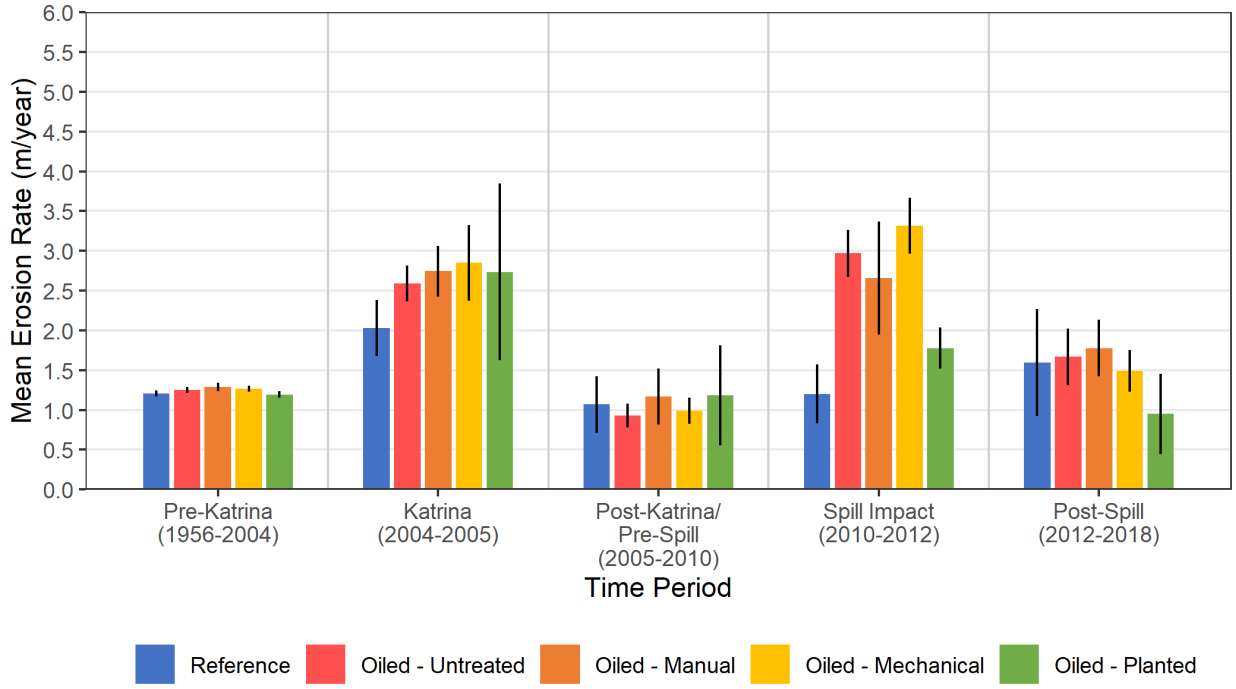
Supplementary Figure



Supplementary Figure S1. Remote sensing measured marsh shoreline erosion rates 1956-2018 (m yr^{-1}), without the influence of Hurricane Isaac in 2012. Data are means with 90% confidence intervals, n= 5 for Reference, 9 for Oiled-Untreated, 5 for Oiled-Manual, 9 for Oiled-Mechanical, and 5 for Oiled-Mechanical-Planted treatments. Marsh erosion rates differed among oiling/treatment categories ($F_{4,28}$ = 9.207, p = 0.000); among time-periods ($F_{2.42,67.77}$ = 62.222, p = 0.000); and for the interaction of oiling/treatment and time-period ($F_{9.68,67.77}$ = 4.518, p = 0.000). In the post-Katrina/pre-spill period (2005-2010) erosion differences were not observed among any oiling/treatment classes, including Reference (p = 0.873 to 1.000). In the spill impact period (2010-2012) erosion differences were observed between: Reference versus all oiled classes (p = 0.000) except Planted (p = 0.324); and between Planted versus all other oiled classes (p = 0.000 to 0.048). In the post-spill period (2012-2018) erosion differences were observed between: Planted versus Manual treatment and Untreated (p = 0.070 and 0.077). Between the post-Katrina/pre-spill and spill impact time-periods, erosion differences were not observed within the Reference (p = 0.992) or Planted (p = 0.276) classes; however, differences were observed between these time-periods within each of the other oiled classes (p = 0.000 in all cases). See Supplementary Table S4 for detailed two-way mixed ANOVA results. Tukey's test was used for all pairwise comparisons after ANOVA.